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### Power semiconductor device, power semiconductor chip including the same, and method for manufacturing the same

#### Abstract

A power semiconductor device includes a plurality of gate electrodes configured to be recessed from a first surface of a semiconductor substrate to a second surface of the semiconductor substrate, the second surface being opposite to the first surface, an emitter region configured to make contact with a trench and the first surface, being provided between respective ones of the plurality of gate electrodes, and including impurities of a first conductive type, a collector region configured to make contact with the second surface, and including second impurities of a second conductive type opposite to the first conductive type, a floating region extending toward the second surface in an extension direction of the trench while surrounding a bottom surface of the trench, and including the second impurities, and a trench emitter region interposed between the plurality of gate electrodes in the trench.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit under 35 USC § 119(a) of Korean Patent Application No. 10-2022-0050744, filed in the Korean Intellectual Property Office on Apr. 25, 2022, the entire contents of which are incorporated herein by reference for all purposes.

### BACKGROUND

#### 1. Field

(2) The present disclosure relates to a power semiconductor device, capable of performing a switching operation, a power semiconductor chip including the same, and a method for manufacturing the same.

#### 2. Description of the Related Art

(3) A power semiconductor device may refer to a semiconductor device capable of performing a switching operation under a higher voltage and higher current environment. The power semiconductor device has been mainly used in a field (e.g., an inverter device) requiring higher power switching (that is, a switching operation under the higher voltage and higher current environment). The power semiconductor device may include, for example, an insulated gate bipolar transistor (IGBT) or a power MOSFET. Such a power semiconductor device basically requires a withstand voltage characteristic for a higher voltage. Recently, a higher-speed switching operation has been additionally required.

#### SUMMARY

(4) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(5) In one general aspect, here is provided a power semiconductor device including a plurality of gate electrodes configured to be recessed from a first surface of a semiconductor substrate to a second surface of the semiconductor substrate, the second surface being opposite to the first surface, an emitter region configured to make contact with a trench and the first surface, wherein the emitter region is provided between respective ones of the plurality of gate electrodes, wherein the emitter region includes first impurities of a first conductive type, a collector region configured to make contact with the second surface, and including second impurities of a second conductive type opposite to the first conductive type, a floating region extending toward the second surface in an extension direction of the trench while surrounding a bottom surface of the trench, and including the second impurities, and a trench emitter region interposed between the plurality of gate electrodes in the trench.

(6) The power semiconductor device may also include a gate insulating film, the gate insulating film being interposed between a side surface of the trench and the plurality of gate electrodes, the gate insulating film having a first thickness that is less than a second thickness of a gate insulating film interposed between the bottom surface of the trench and the plurality of gate electrodes.

(7) A first length of the trench emitter region extending from the first surface of the semiconductor substrate may be longer than a second length of each of the plurality of gate electrodes extending from the first surface.

(8) The trench emitter region may be electrically connected to the emitter region through an emitter electrode.

(9) The trench emitter region may include polysilicon.

(10) A third length of the floating region in the extension direction may be greater than a first width of the floating region.

(11) A first width of the floating region may be greater than a second width of the trench.

(12) A first depth from the first surface to an upper end of the floating region may be shallower than a second depth from the first surface to a lower end of the trench.

(13) A lower end of the floating region may be spaced apart from an upper end of the collector region by a specific distance.

(14) The emitter region may include a plurality of regions disposed at opposite sides of the trench, respectively.

(15) The power semiconductor device may also include a drift region disposed on the collector region, the drift region extending in the extension direction while making contact with a side surface of each of the floating region and the trench and the drift region may include the first impurities.

(16) A first doping concentration of the first impurities in the emitter region may be heavier than a second doping concentration of the first impurities in the drift region.

(17) The plurality of gate electrodes may be disposed in a stripe type or a ring type, when viewed

from a plan view.

(18) In another general aspect, here is provided a power semiconductor chip that includes a cell region including the power semiconductor device and a peripheral region including a gate terminal, an emitter terminal, and a collector terminal electrically connected to the plurality of gate electrodes, the emitter region, and the collector region, respectively.

(19) In yet another general aspect, here is provided a method for manufacturing a power semiconductor device, the method including forming a first drift region including first impurities of a first conductive type, through epitaxial growth, forming a floating region by implanting second impurities including a second conductive type opposite to the first conductive type, into an upper portion of the first drift region, forming a second drift region including the first impurities on the first drift region through the epitaxial growth, forming an emitter region by implanting the first impurities into an upper portion of the second drift region, forming a trench through the emitter region by performing an etching process to etch at least a portion of the floating region, forming a plurality of gate electrodes by depositing a conductive material to be spaced apart from a surface of the trench, and forming a trench emitter region by depositing a conductive material between the plurality of gate electrodes such that the trench emitter region is spaced apart from each of the plurality of gate electrodes.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view illustrating a power semiconductor device, according to an embodiment of the present disclosure;

(2) FIG. 2 is a view schematically illustrating a power semiconductor chip including the power semiconductor device of FIG. 1;

(3) FIG. 3 is a view illustrating a portion of a cell region illustrated in FIG. 2;

(4) FIG. 4 is a view illustrating another example of a portion of a cell region illustrated in FIG. 2;

(5) FIG. 5A is a view illustrating a cross section of a cell region of FIG. 3 or 4;

(6) FIG. 5B is an enlarged view illustrating a partial region of FIG. 5A;

(7) FIG. 5C is a graph illustrating the distribution of an electric field in the structure illustrated in FIG. 5A;

(8) FIG. 5D is a graph illustrating the waveform of an electrical short circuit in the structure illustrated in FIG. 5A; and

(9) FIGS. 6A to 6O are views illustrating a method for manufacturing a power semiconductor device, according to an embodiment of the present disclosure.

(10) Throughout the drawings and the detailed description, unless otherwise described or provided, the same drawing reference numerals will be understood to refer to the same elements, features, and structures. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

### DETAILED DESCRIPTION

(11) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order.

(12) The features described herein may be embodied in different forms and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been

provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.

(13) Advantages and features of the present disclosure and methods of achieving the advantages and features will be clear with reference to embodiments described in detail below together with the accompanying drawings. However, the present disclosure is not limited to the embodiments disclosed herein but will be implemented in various forms. The embodiments of the present disclosure are provided so that the present disclosure is completely disclosed, and a person with ordinary skill in the art can fully understand the scope of the present disclosure. The present disclosure will be defined only by the scope of the appended claims. Meanwhile, the terms used in the present specification are for explaining the embodiments, not for limiting the present disclosure.

(14) Terms, such as first, second, A, B, (a), (b) or the like, may be used herein to describe components. Each of these terminologies is not used to define an essence, order or sequence of a corresponding component but used merely to distinguish the corresponding component from other component(s). For example, a first component may be referred to as a second component, and similarly the second component may also be referred to as the first component.

(15) Throughout the specification, when a component is described as being “connected to,” or “coupled to” another component, it may be directly “connected to,” or “coupled to” the other component, or there may be one or more other components intervening therebetween. In contrast, when an element is described as being “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.

(16) In a description of the embodiment, in a case in which any one element is described as being formed on or under another element, such a description includes both a case in which the two elements are formed in direct contact with each other and a case in which the two elements are in indirect contact with each other with one or more other elements interposed between the two elements. In addition, when one element is described as being formed on or under another element, such a description may include a case in which the one element is formed at an upper side or a lower side with respect to another element.

(17) The singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises/comprising” and/or “includes/including” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.

(18) FIG. 1 is a view illustrating a power semiconductor device, according to an embodiment of the present disclosure.

(19) Referring to FIG. 1, a power semiconductor device **10** according to an embodiment of the present disclosure, which is a device operated by a switch under a higher voltage and a higher current, may be employed for an inverter.

(20) As illustrated in FIG. 1, the power semiconductor device **10** may include an insulated gate bipolar transistor (IGBT). According to another embodiment, although the power semiconductor device **10** may be realized in another type of power transistor (e.g., a metal oxide semiconductor field effect transistor (MOSFET) based on a silicon carbide). The technical spirit of the present disclosure will be described on the assumption that the power semiconductor device **10** is realized.

(21) The IGBT may have a structure in which a gate of MOSFET is coupled to an emitter and a collector of a bipolar transistor. Accordingly, the power semiconductor device **10** may have both the withstand voltage characteristic of the bipolar transistor and the more rapid switching speed of the MOSFET.

(22) The IGBT may be a self-quenching device in which the gate-emitter voltage  $V_{ge}$  is changed to

be turned on or off depending on the driving voltage applied to the gate. In this case, the self-quenching type device is a device to receive an input signal without depending on the outside, thereby allowing a current to flow between two terminals (that is, an emitter and a collector) or preventing the current from flowing between the two terminals through self-control ability.

(23) The IGBT may include a gate terminal G connected to a gate, an emitter terminal E connected to the emitter, and a collector terminal C connected to the collector. The IGBT is turned on or off in response to a driving voltage applied to the gate terminal G, thereby allowing the current to flow between two terminals (that is, an emitter and a collector) or preventing the current from flowing between the two terminals. The gate terminal G may be connected a driver circuit (not illustrated) to transfer the driving voltage, and the emitter terminal E and the collector terminal C may be connected to a load (not illustrated) to provide a switching function.

(24) According to an embodiment, the power semiconductor device **10** may further include a flyback diode Df connected between the emitter terminal E and the collector terminal C. The flyback diode Df may provide a loop that allows carriers (holes or electrons), which are accumulated in the emitter or collector due to the on-operation of the IGBT, to be re-emitted in the off-operation of the IGBT. The flyback diode Df may prevent an impulse voltage by providing such a loop, thereby preventing the damage to a load connected to the IGBT or the power semiconductor device **10**.

(25) FIG. **2** is a view schematically illustrating a power semiconductor chip including the power semiconductor device of FIG. **1**.

(26) Referring to FIG. **2**, a power semiconductor chip **20** may be a semiconductor chip including the power semiconductor device **10** of FIG. **1**. The power semiconductor chip **20** may include one power semiconductor device **10** formed in an integral type, or may include a plurality of power semiconductor devices **10** according to another embodiment.

(27) The power semiconductor chip **20** may include a cell region **100** and a peripheral region **200**.

(28) The cell region **100** may include at least one power semiconductor device **10**. For example, the arrangement form of the power semiconductor device **10** may correspond to a stripe type as illustrated in FIG. **3** or a closed type as illustrated in FIG. **4**. Such an arrangement form will be described later with reference to FIGS. **3** and **4**.

(29) The cell region **100** may further include a device for monitoring the state of at least one power semiconductor device **10**.

(30) According to an embodiment, the cell region **100** may further include a current sensing transistor substantially the same as the power semiconductor device **10** and having a structure reduced at a predetermined ratio. For example, the current sensing transistor may be connected in parallel to the IGBT of the power semiconductor device **10**. In this case, a specific resistor may be connected between one terminal of the current sensing transistor and one terminal of the IGBT in the parallel connection. An output current of the current sensing transistor (e.g., the output current from the emitter terminal or collector terminal) may have a ratio associated with a specific ratio to the output current from the power semiconductor device **10**, and the output current from the current sensing transistor may be used to indirectly monitor the state of the power semiconductor device **10**.

(31) According to another embodiment, the cell region **100** may further include a temperature sensor disposed adjacent to the power semiconductor device **10** to sense the temperature of the power semiconductor device **10**. For example, the temperature sensor may include a junction diode which outputs a current varying depending on a temperature. The current of the junction diode may be used to monitor the temperature of the power semiconductor device **10**.

(32) The peripheral region **200** may be disposed outside the cell region **100** while surrounding the cell region **100**. The peripheral region **200** may include a plurality of terminals for electrical connection between an external circuit (e.g., a driving circuit, a load, or a test circuit) and the cell region **100**. The plurality of terminals may include a gate terminal G, an emitter terminal E, and a

collector terminal C of the IGBT. In addition, the plurality of terminals may include each terminal of the current sensing transistor and/or the temperature sensor that may be included in the cell region **100**.

(33) FIG. **3** is a view illustrating a portion of a cell region illustrated in FIG. **2**.

(34) Referring to FIG. **3**, a part **300** of the cell region **100** of FIG. **2** is illustrated. The cell region **100** may have the form in which the part **300** of the cell region **100** is repeatedly arranged in a vertical direction or a horizontal direction, when employing the part **300** of the cell region **100** as one unit, but the present disclosure is not limited thereto.

(35) The part **300** of the cell region **100** may include a gate electrode **120**, a gate insulating film **122**, an emitter region **130**, an emitter electrode **132**, and a trench emitter region **135**.

(36) The gate electrode **120** corresponds to the gate of the IGBT in FIG. **1**, and the gate insulating film **122** may electrically separate the gate electrode **120** from another component. The emitter region **130** may correspond to the emitter electrode **132**, and the trench emitter region **135** may correspond to the emitter of the IGBT in FIG. **1**.

(37) The part **300** of the cell region **100** represents a plane positioned at a specific height when viewed in a vertical cross section of the cell region **100**, and a component corresponding to the collector of the IGBT in FIG. **1** may be positioned at a height different from that of the part **300** of the cell region **100**.

(38) Meanwhile, the detailed structure, the function, and the material of each of the gate electrode **120**, the gate insulating film **122**, the emitter region **130**, the emitter electrode **132**, and the trench emitter region **135** will be described later with reference to FIG. **5A**. Referring to FIG. **3**, the arrangement form of the gate electrode **120**, the gate insulating film **122**, the emitter region **130**, the emitter electrode **132**, and the trench emitter region **135** when viewed on a plan view will be described.

(39) The gate electrode **120** may extend in one direction (the vertical direction in FIG. **3**). The gate insulating film **122** may extend in parallel to the gate electrode **120** in one direction while making contact with the left or right side of the gate electrode **120**.

(40) The emitter region **130** may extend in parallel to the gate electrode **120** in one direction, while making contact with the gate insulating film **122** disposed on the left side of the gate electrode **120** or the gate insulating film **122** disposed on the right side of the gate electrode **120**. In other words, the gate insulating film **122** may be interposed between the gate electrode **120** and the emitter region **130**, and the gate electrode **120** and the emitter region **130** may be electrically isolated from each other by the gate insulating film **122**.

(41) The trench emitter region **135** may be disposed inside the adjacent gate electrodes **120** while being spaced apart from each of the gate electrodes **120**, and the gate insulating film **122** may be interposed between each of the gate electrodes **120** and the trench emitter region **135** to electrically isolate each gate electrode **120** from the trench emitter region **135**. The trench emitter region **135** may extend in parallel to each of the gate electrodes **120**.

(42) Unit cells, each of which includes the trench emitter region **135**, the gate electrodes **120** disposed on opposite sides of the trench emitter region **135**, the emitter regions **130** disposed on the left and right sides of gate electrode **120**, and the gate insulating films **122** to electrically isolate the components from each other, may be arranged while being spaced apart from each other in another direction perpendicular to one direction (in the horizontal direction in FIG. **3**). As illustrated in FIG. **3**, emitter electrodes **132** may be disposed at the left and right sides of the unit cell, which is positioned at the left side, to make contact with the unit cell. As illustrated in FIG. **3**, emitter electrodes **132** may be disposed at the left and right sides of the unit cell, which is positioned at the right side, to make contact with the unit cell. Accordingly, the emitter electrode **132** may be interposed between unit cells spaced apart from each other in the another direction.

(43) Since the unit cells has a stripe shape as in FIG. **3**, the arrangement form of the power semiconductor device **10** may be defined as a stripe type.

(44) FIG. 4 is a view illustrating another example of a portion of a cell region illustrated in FIG. 2.

(45) Referring to FIG. 4, a part **400** of the cell region **100** of FIG. 2 is illustrated. The cell region **100** may have the form in which the part **400** of the cell region **100** is repeatedly arranged in a vertical direction or a horizontal direction, when employing the part **400** of the cell region **100** as one unit, but the present disclosure is not limited thereto.

(46) Since the part **400** of the cell region **100** is substantially the same as the part **300** of the cell region **100** described in FIG. 3, except for an arrangement form, the duplication thereof will be omitted to avoid redundancy.

(47) The gate electrode **120** may have a rectangular ring shape. The gate insulating film **122** may extend in parallel to the gate electrode **120** while making contact with one side of the gate electrode **120**.

(48) The emitter region **130** may extend in parallel to the gate electrode **120**, while making contact with the gate insulating film **122** disposed on one side of the gate electrode **120** or the gate insulating film **122** disposed on an opposite side of the gate electrode **120**. In other words, the gate insulating film **122** may be interposed between the gate electrode **120** and the emitter region **130**, and the gate electrode **120** and the emitter region **130** may be electrically isolated from each other by the gate insulating film **122**.

(49) The trench emitter region **135** may be interposed between the adjacent gate electrodes **120** while being spaced apart from each of the gate electrodes **120**, and the gate insulating film **122** may be interposed between each of the gate electrodes **120** and the trench emitter region **135** to electrically isolate each gate electrode **120** from the trench emitter region **135**. The trench emitter region **135** may extend in parallel to each of the gate electrodes **120**.

(50) Unit cells, each of which includes a trench emitter region **135**, gate electrodes **120** surrounded by the trench emitter region **135** or surrounding the trench emitter region **135**, emitter regions **130** surrounded by the gate electrodes **120** or surrounding the gate electrodes **120**, and gate insulating films **122** to electrically isolate the above components from each other may substantially have a rectangular ring shape.

(51) As illustrated in FIG. 4, emitter electrodes **132** may be disposed at the left and right sides of the unit cell to make contact with the unit cell.

(52) Since the unit cell has a ring shape as in FIG. 4, the arrangement form of the power semiconductor device **10** may be defined as a closed type.

(53) The stripe type illustrated in FIG. 3 may be realized at a higher channel density (the density of a channel between the emitter and collector), which is advantageous in terms of a static characteristic of the IGBT. The closed type illustrated in FIG. 4 may more easily adjust the size and the ratio of a parasitic capacitance, which is advantageous in terms of switching stability of the IGBT. The stripe type or the closed type may be selected depending on the use and the requirements of the IGBT.

(54) According to another embodiment, the cell region **100** may be realized in a ladder type which is a merge type of the stripe type and the closed type. The ladder type may refer to a form in which some of the gate electrodes **120** are spaced apart from each other by a specific distance while extending in one direction, which is similar to a stripe type, and others of the gate electrodes **120** extend in a direction perpendicular to the one direction to connect the some of the gate electrodes **120** belonging the stripe type. The ladder type may have the stripe type and the closed type.

(55) FIG. 5A is a view illustrating a cross section of a cell region of FIG. 3 or 4; FIG. 5B is an enlarged view illustrating a partial region of FIG. 5A. FIG. 5C is a graph illustrating the distribution of an electric field in the structure illustrated in FIG. 5A. FIG. 5D is a graph illustrating the waveform of an electrical short circuit in the structure illustrated in FIG. 5A.

(56) FIG. 5A illustrates a cross section **500** of the cell region **100** taken along first line A-A' of FIG. 3 or second line B-B' of FIG. 4, and the first line A-A' or the second line B-B' may correspond to third line C-C' of FIG. 5A.



(57) The cross section **500** may include a semiconductor substrate **110**, a collector electrode **112** disposed at a lower portion of the semiconductor substrate **110**, and the gate insulating film **122** and the emitter electrode **132** having at least a portion protruding from an upper portion of the semiconductor substrate **110**

(58) The semiconductor substrate **110** may refer to at least one semiconductor material layer (for example, an epitaxial layer). For example, the semiconductor substrate **110** may include a semiconductor material as silicon, germanium, or a silicon-germanium-based semiconductor material. The semiconductor substrate **110** may refer to a region from the third line C-C' to a lower end of the collector electrode **112**.

(59) The semiconductor substrate **110** may include a collector region **114**, a drift region **116**, a well region **118**, a gate electrode **120**, a portion of a gate insulating film **122**, an emitter region **130**, a portion of an emitter electrode **132**, a trench emitter region **135**, and a floating region **150**. The semiconductor substrate **110** may include a top surface (or the first surface) and a bottom surface (or the second surface).

(60) The collector region **114** may include impurities in a second conductive type. The second conductive type may be a P type or an N type, and may be an opposite conductive type of a first conductive type. The collector region **114** may be formed to have a specific thickness while making contact with the bottom surface of the semiconductor substrate **110**. The collector region **114** may be disposed to make contact with the collector electrode **112** under the collector region **114**. The collector electrode **112** may be electrically connected to the collector terminal C of FIG. 1, and may include at least one of polysilicon, metal, a metal nitride, or a metal silicide.

(61) The collector region **114** and the collector electrode **112** may correspond to the collector of the IGBT of FIG. 1.

(62) The drift region **116** may include impurities in the first conductive type. The first conductive type may be a P type or an N type, and may be an opposite conductive type of the second conductive type. The drift region **116** may provide a vertical movement path of charges (holes and electrons) between the collector region **114** and the emitter region **130**. The drift region **116** is disposed under the collector region **114** and may extend in the extension direction of a trench TH (the vertical direction in FIG. 5A) while making contact with the side surfaces of each of the floating region **150** and the trench TH.

(63) The well region **118** may include impurities in the second conductive type. The well region **118** may be disposed between the drift region **116** and the emitter region **130** and between the drift region **116** and the emitter electrode **132**. In addition, the well region **118** may be interposed between adjacent trenches TH provided therein with the gate electrode **120** and the gate insulating film **122**. The well region **118** may be disposed to make contact with the drift region **107**, the gate insulating film **122**, the emitter region **130**, and the emitter electrode **132**. The trench TH may refer to a structure recessed to a specific depth from one surface of the semiconductor substrate **110** toward an opposite surface of the semiconductor substrate **110**.

(64) The depth from the top surface of the semiconductor substrate **110** to a bottom surface of the well region **118** may be shallower than the depth from the top surface of the semiconductor substrate **110** to a bottom surface of the trench TH.

(65) The gate electrode **120** may include a material having conductivity and may include at least one of polysilicon, metal, a metal nitride, or a metal silicide. The gate electrode **120** may be disposed to fill the inside of the trench TH while making contact with the gate insulating film **122**. Accordingly, the gate electrode **120** may have a shape recessed to a specific depth from the top surface of the semiconductor substrate **110** toward the bottom, surface of the semiconductor substrate **110**.

(66) The gate electrode **120** may include two parts separated from each other in the trench TH. In other words, the trench emitter region **135** may be disposed at the center of the trench TH, and the gate electrodes **120** may be disposed on left and right sides of the trench emitter region **135**.

(67) The gate electrode **120** may extend in parallel to the trench emitter region **135** from the top surface of the semiconductor substrate **110** toward the bottom surface of the semiconductor substrate **110**. The length of the gate electrode **120** extending from the top surface of the semiconductor substrate **110** may be shorter than the length of the trench emitter region **135** extending from the top surface of the semiconductor substrate **110**, but the scope of the present disclosure is not limited thereto.

(68) The gate electrode **120** may correspond to a gate of the IGBT of FIG. **1** and may be electrically connected to the gate terminal G of FIG. **1**.

(69) The gate insulating film **122** may include an insulating material, for example, at least one of an insulating material, for example, a silicon oxide, a silicon nitride, a germanium oxide, a germanium nitride, a hafnium oxide, a zirconium oxide, or an aluminum oxide. The gate insulating film **122** may be disposed to make contact with the gate electrode **120** inside the trench TH, while extending along the side surface of the trench TH. In addition, the gate insulating film **122** may extend to the outside of the semiconductor substrate **110** and be disposed above the gate electrode **120**, the emitter region **130**, and the trench emitter region **135**. The gate insulating film **122** extending to the outside of the semiconductor substrate **110** may be referred to as an interlayer insulating layer.

(70) The gate insulating film **122** positioned inside the trench TH may be interposed between the surface of the trench TH and the gate electrode **120**, between the gate electrode **120** a region (that is, an upper region) of the trench emitter region **135**, and between the gate electrode **120** and the trench emitter region **135**.

(71) The gate insulating film **122** may electrically isolate the gate electrode **120** from each of the drift region **116**, the well region **118**, the emitter region **130**, the emitter electrode **132** and the floating region **150**. In addition, the gate insulating film **122** may electrically isolate the gate electrode **120** and the trench emitter region **135** from each other.

(72) The thickness of the gate insulating film **122** interposed between the side surface of the trench TH and the gate electrode **120** may be thinner than the thickness of the gate insulating film **122** positioned between the bottom surface of the trench TH and the gate electrode **120**.

(73) The well region **130** may include impurities in the first conductive type. The doping concentration of impurities in the first conductive type in the emitter region **130** may be heavier than the doping concentration of impurities in the first conductivity type in the drift region **116**. The emitter region **130** may make contact with the top surface of the semiconductor substrate **110**, and may be interposed between the gate insulating film **122** in the trench TH and the emitter electrode **132** while making contact with each of the gate insulating film **122** (or the trench TH) and the emitter electrode **132**. In addition, the bottom surface of the emitter region **130** may be disposed to make contact with the well region **118**. The emitter region **130** may be disposed on one side and the opposite side (or both left and right sides) of the trench TH, respectively.

(74) The gate electrode **132** may include a material having conductivity and may include at least one of polysilicon, metal, a metal nitride, or a metal silicide. The emitter electrode **132** may be disposed to cover the upper portion of the semiconductor substrate **110** while making contact with the gate insulating film **122** protruding outside the semiconductor substrate **110**, the well region **118**, and the emitter region **130**.

(75) The emitter electrode **132** may apply a voltage, which is lower than a voltage across a collector of the IGBT, to the well region **110** by serving as a contact to the well region **118**. Regarding the emitter region **130**, the emitter electrode **132** may serve as an electrode to electrically connect the emitter region **130** to the emitter terminal E of FIG. **1**. The collector region **130** and the collector electrode **132** may correspond to the emitter of the IGBT of FIG. **1**.

(76) The trench emitter region **135** may include a material having conductivity and may include at least one of polysilicon, metal, a metal nitride, or a metal silicide. The trench emitter region **135** may be interposed between the gate electrodes **120**. In addition, the trench emitter region **135** may be disposed to make contact with the gate insulating film **122** to be electrically isolated from the

surfaces of the gate electrodes **120** and the trench TH.

(77) The gate electrode **135** may extend in parallel to the gate electrode **120** from the top surface of the semiconductor substrate **110** toward the bottom surface of the semiconductor substrate **110**. The length of the trench emitter region **135** extending from the top surface of the semiconductor substrate **110** may be longer than the length of the gate electrode **120** extending from the top surface of the semiconductor substrate **110**, but the scope of the present disclosure is not limited thereto.

(78) The trench emitter region **135** may be electrically connected to the emitter electrode **132**, and may correspond to the emitter of the IGBT of FIG. **1** together with the emitter region **130** electrically connected to the emitter electrode **132**. As a contact hole is formed at an arbitrary (not illustrated) of the cell region **100** or the peripheral region **200**, by removing at least a portion of the gate insulating film **122** at the upper portion of the trench emitter region **135**, the trench emitter region **135** may be electrically connected to the emitter electrode **132** through the contact hole, but the present disclosure is not limited thereto,

(79) The floating region **150** may include impurities in the second conductive type. The floating region **150** may be an electrically floating region having bias voltage applied thereto from the outside.

(80) The floating region **150** may extend toward the collector region **114** in the extension direction (the vertical direction of FIG. **5A**) of the trench TH while surrounding the bottom surface of the trench TH.

(81) The floating region **150** may have a width **W** wider than a width **W'** of the trench TH to surround the entire bottom surface of the trench TH. When the floating region **150** is not included, an electric field may be concentrated on the bottom surface of the trench TH during operation of the IGBT, such that the maximum electric field may be formed on the bottom surface of the trench (TH). Since the thickness of the gate insulating film **122** is only about several hundred nm, the breakdown voltage of the IGBT may be decreased, as the electric field is concentrated on the bottom surface of the trench TH. As the breakdown voltage is decreased, the normal switching operation of the IGBT may be difficult. However, according to the present disclosure, when the floating region **150** surrounds the entire portion of the bottom surface of the trench TH, the electric field is distributed into the floating region **150** instead of being concentrated on the bottom surface of the trench TH due to a charge sharing phenomenon. Accordingly, the maximum electric field may be formed on the bottom surface of the floating region **150**. Accordingly, the breakdown voltage of the IGBT is not decreased, but the switching performance of the IGBT may be ensured. In other words, since the IGBT may perform the switching operation under the higher voltage environment due to the floating region **150**, the withstand voltage characteristic may be improved. In addition, since the gate insulating film **122** positioned under the gate electrode **120** has a thickness thicker than that of the gate insulating film **122** positioned on the side surface of the gate electrode **120**, the electric field may be prevented from being concentrated on the bottom surface of the trench TH.

(82) As the position of the maximum electric field is moved due to the floating region **150** and the internal structure of the trench TH, the margin for the breakdown voltage of the IGBT may be ensured. Accordingly, the doping concentration of impurities in the first conductive type in the drift region **116** may be set to be heavier to increase the density of a carrier (e.g., a hole) when the IGBT is turned on, such that the switching performance is improved. Accordingly, the obtained effect may be the substantially same as that the spacing between mutually adjacent trenches TH may be reduced to increase the channel density.

(83) FIG. **5C** illustrates an electric field distributed line D-D' illustrated in FIG. **5A**. A first electric field EF\_P is one example of an electric field distributed in line D-D' marked in FIG. **5A**. A second electric field EF\_C is an example of an electric field at a position corresponding to line D-D' in a structure (that is, a gate electrode is filled in the trench TH without the trench emitter region **135**,

and the floating region **150** is omitted, according to a comparative example of the present disclosure) without the emitter region **130** and the floating region **150**, which are provided in the structure of FIG. 5A.

(84) As the electric field is distributed due to the charge sharing phenomenon by the floating region **150**, and the thicker thickness of the gate insulating film **122** positioned to be adjacent to the bottom surface of the trench TH, the first electric field EF\_P may have the intensity of the electric field, which is substantially greater than the intensity of the second electric field EF\_C.

Accordingly, the area of the first electric field EF\_P may be greater than the area of the second electric field EF\_C. In this case, the area of the first electric field EF\_P or the second electric field EF\_C may refer to a value obtained by integrating the intensity of the electric field along line D-D'. In addition, the area of the first electric field EF\_P or the area of the second electric field EF\_C may indicate the intensity of the breakdown voltage of the IGBT. The breakdown voltage of the IGBT having the structure of FIG. 5A according to an embodiment of the present disclosure may be greater than the breakdown voltage of the IGBT having the structure according to a comparative example of the present disclosure.

(85) Referring back to FIG. 5A, the bottom surface of the trench TH may have a substantially rounded shape such that an angled region is not present at a part extending from the bottom surface of the trench TH to the side surface of the trench TH. When the angled region is present under the trench TH, a breakdown voltage of the IGBT may be decreased, as an electric field is concentrated in the angled region. Accordingly, the trench TH according to the present disclosure may have a bottom surface having the substantially rounded shape to improve the withstand voltage characteristic.

(86) The width W of the floating region **150** may be determined to be greater than the width W' of the trench TH with a difference of only a degree to surround the entire bottom surface of the trench TH. In other words, the width W of the floating region **150** may have a value significantly approximate to the width W' of the trench TH. Accordingly, one unit including the trench TH and the floating region **150** surrounding the lower portion of the trench TH may have a narrower width W, and may be reduced in pitch between mutually different gate electrodes **120**, thereby increasing the element density of the cell region **100**.

(87) The collector-emitter charge movement path is formed in the drift region **116** between the gate electrodes **120** adjacent to each other and between the floating regions **150** adjacent to each other, when IGBT is operated. When the spacing between the gate electrodes **120** adjacent to each other and the spacing between the floating regions **150** adjacent to each other is reduced, the channel density between the collector and the emitter is increased to increase the gain of the IGBT, and the carrier movement between the collector and the emitter is made easy to reduce the JFET resistance, thereby improving the switching performance. However, when the spacing between the gate electrodes **120** adjacent to each other and the spacing between the floating regions **150** is excessively reduced, the area of the charge movement path is reduced to increase the turn-on resistance of the IGBT. In particular, when the spacing between the floating regions **150** adjacent to each other is excessively reduced, the area of the charge movement path is more narrowed due to the depletion region formed due to the junction between the floating region **150** and the drift region **116**. The increase in the turn-on resistance (the ratio of a voltage Vce to a current Ice between the emitter and the collector) of the IGBT may lower the switching speed of the IGBT. Accordingly, the doping concentration of the impurities in the first conductive type in the drift region **116** (a drift region above a virtual line extending through a lower end of the floating region **150** and parallel to one surface of the semiconductor substrate **110**) between the gate electrodes **120** adjacent to each other and between the floating regions **150** adjacent to each other is relatively increased the doping concentration of impurities in the first conductive type in the drift region **116** (that is, a drift region under the virtual line extending through the lower end of the floating region **150** and parallel to one surface of the semiconductor substrate **110**) under the floating regions **150**, thereby reducing the

turn-on resistance of the IGBT.

(88) The floating region **150** may have a length  $L$  extending from an upper end of the floating region **150** to a lower end of the floating region **150** in the extension direction of the trench **TH**. When viewed from the top surface of the semiconductor substrate **110**, the depth to the upper end of the floating region **150** is shallower than the depth to the lower end of the trench **TH**, and deeper than the depth to the lower end of the well region **118**. In addition, when viewed from the bottom surface of the semiconductor substrate **110**, the depth to the lower end of the floating region **150** may be deeper than the depth to the upper end of the collector region **114**. In other words, the lower end of the floating region **150** and the upper end of the collector region **114** may be spaced apart from each other by a specific distance.

(89) The length  $L$  of the floating region **150** may be greater than the width  $W$  of the floating region **150**, and may be determined to have a value as large as possible. As the floating region **150** is formed with a length as long as possible, the position for the maximum electric field may be adjusted to be as close as possible to the collector region **114**. Accordingly, the breakdown voltage is increased to improve the withstand voltage characteristic.

(90) Referring back to FIG. 5B, an enlarged view of a partial region **550** of FIG. 5A is illustrated. A gate-emitter first parasitic capacitance  $C_{ge1}$  may be made at the interface between the gate electrode **120** and the well region **118** disposed at one side (a left side or a right side) of the gate electrode **120**. A gate-emitter second parasitic capacitance  $C_{ge2}$  may be made on the interface between the gate electrode **120** and the trench emitter region **135** disposed at one side (a left side or a right side) of the gate electrode **120**. In addition, a gate-collector parasitic capacitance  $C_{gc}$  may be made at an interface between the gate electrode **120** and the drift regions **116** disposed on the left and right sides of the gate electrode **120**.

(91) The total gate-emitter parasitic capacitance  $C_{ge}$  in the IGBT may be the sum of the first parasitic capacitance  $C_{ge1}$  made between the gate electrode **120** and the emitter region **130** and the second parasitic capacitance  $C_{ge2}$  made between the gate electrode **120** and the trench emitter region **135**. In other words, since the trench emitter region **135** is present inside the trench **TH**, a significantly higher gate-emitter parasitic capacitance  $C_{ge}$  (two times of a capacitance in the case that only the emitter region **130** is present) may be made in the trench **TH**, as compared to the case that only the emitter region **130** is present.

(92) As the gate-emitter parasitic capacitance  $C_{ge}$  is increased, the switching operation stability may be increased, but the switching speed may be decreased. Accordingly, the thickness of the gate insulating film **122**, the doping concentration of the well region **118** and the emitter region **130**, and the shape/material of the trench emitter region **135** may be adjusted such that the gate-emitter parasitic capacitance  $C_{ge}$  has an appropriate value

(93) As illustrated in FIG. 5A, the gate insulating films **122** interposed between the gate electrode **120** and the well region **118**, and between the gate electrode **120** and the trench emitter region **135** have thinner thicknesses, such that the gate-emitter parasitic capacitance  $C_{ge}$  has a greater value.

(94) The total gate-collector parasitic capacitance  $C_{gc}$  in the IGBT may be the sum of a parasitic capacitance made at the interface between the gate electrode **120** and the drift regions **116** positioned at the left and right sides, and a parasitic capacitance generated at the interface between the bottom surface of the floating region **150** and the drift region **116**. The parasitic capacitance may be made in another region. However, the following description will be made while focusing on the main parasitic capacitance. In this case, the size of the parasitic capacitance generated at the interface between the bottom surface of the floating region **150** and the drift region **116** may be decreased, as the distance from the bottom surface of the gate electrode **120** to the bottom surface of the floating area **150** is increased. In other words, on the assumption that the upper depth of the floating region **150** is fixed, the gate-collector parasitic capacitance  $C_{gc}$  of the IGBT may be decreased, as the length  $L$  of the floating region **150** is increased.

(95) In addition, the sizes of the parasitic capacitances made at the interface between the gate

electrode **120** and the left and right drift regions **116** may be decreased, as the thickness of the gate insulating film **122** disposed under the gate electrode **120** is increased. In other words, as the length of the gate electrode **120** extending from the top surface of the semiconductor substrate **110** toward the bottom surface of the semiconductor substrate **110** is decreased, the size of the parasitic capacitance made at the interface between the gate electrode **120** and the left and right drift regions **116** may be decreased. For example, the distance between an arbitrary point of the drift region **116** adjacent to the upper end of the floating region **150** and the gate electrode **120** may be increased, as the length of the gate electrode **120** extending from the top surface of the semiconductor substrate **110** toward the bottom surface of the semiconductor substrate **110** is decreased, such that the size of the parasitic capacitance is reduced. In addition, the gate electrode **120** has a form in which the width of the gate electrode **120** is gradually reduced toward the lower end of the gate electrode **120**, and the thickness of the gate insulating film **122** interposed between the gate electrode **120** and the drift region **116** may be increased. As the thickness of the gate insulating film **122** is increased, the size of the parasitic capacitance made at the interface between the gate electrode **120** and the left and right drift regions **116** may be gradually decreased. In addition, the gate electrode **120** has a shape gradually inclined to the trench emitter region **135** toward the lower end of the gate electrode **120**, and the trench emitter region **135** has a length longer than that of the gate electrode **120**, such that the gate-emitter parasitic capacitance is more increased.

(96) Accordingly, the gate-collector parasitic capacitance  $C_{gc}$  may be adjusted by adjusting at least one of the depth of the floating region **150**, the shape of the gate electrode **120**, or the thickness of the gate insulating film.

(97) In the power semiconductor device having the trench-type gate structure as illustrated in FIG. 5A, when carriers (e.g., holes) are excessively accumulated at an interface of the trench TH, a negative gate capacitance (NGC) phenomenon may occur and a displacement current may be generated. The displacement current may degrade the switching operation stability of the IGBT and has a characteristic proportional to the size of the gate-collector parasitic capacitance  $C_{gc}$ .

(98) According to the present disclosure, the gate-collector parasitic capacitance  $C_{gc}$  may be adjusted by adjusting at least one of the depth of the floating region **150**, the shape of the gate electrode **120**, or the thickness of the gate insulating film, thereby satisfying the necessary switching operation stability.

(99) The switching loss and switching operation stability of the IGBT may depend on the capacitance ratio  $C_{ge}/C_{gc}$  between the gate-emitter parasitic capacitance  $C_{ge}$  and the gate-collector parasitic capacitance  $C_{gc}$ . As the capacitance ratio  $C_{ge}/C_{gc}$  is increased, the switching loss may be reduced, and the switching operation stability may be increased.

(100) The turn-on resistance, which is the ratio of a collector-emitter voltage  $V_{ce}$  to a collector-emitter current  $I_{ce}$ , may indicate the size of the switching loss of the IGBT when the IGBT is turned on. In other words, as the turn-on resistance is reduced, the switching loss of the IGBT may be decreased. In addition, the turn-on resistance may depend on the capacitance ratio  $C_{ge}/C_{gc}$ . In detail, the turn-on resistance may be reduced in the structure, such as the structure of FIG. 5A, of relatively increasing the capacitance ratio  $C_{ge}/C_{gc}$ , and the turn-on resistance may be decreased, thereby reducing the switching loss of the IGBT.

(101) Referring to FIG. 5D, the overshooting degree, which is a characteristic in a short circuit state of the IGBT, of the gate-emitter voltage  $V_{ge}$  in turning on the IGBT may indicate the switching operation stability of the IGBT. In other words, as the overshooting degree of the gate-emitter voltage  $V_{ge}$  is decreased, the switching operation stability may be increased. FIG. 5D illustrates of a simulation result of the gate-emitter voltage  $V_{ge\_P}$  and the short circuit current  $I_{sc\_P}$  over time in the structure (See FIG. 5A) according to an embodiment of the present disclosure, and a simulation result of the gate-emitter voltage  $V_{ge\_C}$  and the short circuit current  $I_{sc\_C}$  in a structure (the structure without the trench emitter region **135** and the floating region **150** provided in the structure of FIG. 5A) according to a comparative example of the present disclosure.

over time, under the same condition (e.g.,  $V_{ge}$  (gate-emitter voltage)=15 V,  $V_{ce}$  (collector-emitter voltage)=200 V, and  $R_{on}$  (turn-on resistance)=1 ohm).

(102) Although the structure according to an embodiment of the present disclosure is tuned with a higher short circuit current as compared to the structure according to the comparative example of the present disclosure ( $I_{sc\_P}$ =8000 A, and  $I_{sc\_C}$ =4400 A), the peak value of  $V_{ge\_P}$  may be about 15.3 V and the peak value of  $V_{ge\_C}$  may be about 15.9 V. In other words, the structure (about 0.3 V) according to an embodiment of the present disclosure may have the overshooting degree of the gate-emitter voltage, which less than that of the structure (about 0.9 V) according to the comparative example of the present disclosure. In addition, the peak value of  $I_{sc\_P}$  may be about 8200 A and the peak value of  $I_{sc\_C}$  may be about 4800 A. In other words, the structure (about 200 A) according to an embodiment of the present disclosure may have the overshooting degree of the short-circuit current, which less than that of the structure (about 400 A) according to the comparative example of the present disclosure.

(103) The overshooting degree of the gate-emitter voltage  $V_{ge}$  may depend on the capacitance ratio  $C_{ge}/C_{gc}$ . According to the structure, such as the structure of FIG. 5A, of relatively increasing the capacitance ratio  $C_{ge}/C_{gc}$ , the overshooting degree of the gate-emitter voltage  $V_{ge}$  may be decreased. Accordingly, the switching operation stability of the IGBT may be increased.

(104) FIGS. 6A to 60 are views illustrating a method for manufacturing a power semiconductor device, according to an embodiment of the present disclosure.

(105) Referring to FIGS. 6A to 60, a structure in which a drift region 116' is stacked on the collector electrode 112 and the collector region 114 may be provided in S10. In this case, the drift region 116' may have a height lower than that of the drift region 116 illustrated in FIG. 5A. The drift region 116' may be formed through epitaxial growth. According to the present disclosure, although the description of S10 has been made on the assumption that the collector electrode 112 and the collector region 114 are already formed under the drift region 116', at least one of the collector electrode 112 and the collector region 114 may be formed after S10.

(106) In 520, a photoresist pattern defining the floating region 150 may be disposed at an upper portion of the drift region 116', and impurities in the second conductive type may be implanted into the upper portion of the drift region 116' to form the floating region 150. The energy for implanting impurities in the second conductive type may be determined such that the length L of the floating region 150 satisfies a specific capacitance ratio  $C_{ge}/C_{gc}$ .

(107) In S30, the drift region 116 may be formed by adding an epitaxial layer in a first conductive type by performing epitaxial growth with respect to the upper portion (that is, the upper portion of a boundary BD) of the drift region 116'. In other words, the drift region 116 may include a drift region 116' (or the first drift region positioned under the boundary BD) formed before forming the floating region 150, and a drift region (or the second drift region positioned on the boundary BD) formed after forming the floating region 150. According to the present disclosure, the floating region 150 positioned inside the drift region 116 is not formed by performing an etching process, a photoresist process, or a polishing process after first forming the entire portion of the drift region 116, but may be formed at a depth, which may not be made by a general implanting device, by performing an implantation process for the floating region 150 in the middle of the epitaxial growth process for the drift region 116.

(108) In S40, after forming the well region 118 by implanting impurities, which is in the second conductive type, into the entire upper portion of the drift region 116 to form the well region 118, a photoresist pattern for defining the emitter region 130 may be disposed at the upper portion of the drift region 116, and the impurities in the first conductive type is implanted into the upper portion of the drift region 116, thereby forming the emitter region 130.

(109) In S50, an etching mask to define the trench TH may be disposed on the well region 118 and the emitter region 130, and the trench TH may be formed through the well region 118 and the emitter region 130 by performing the etching process. The trench TH may be formed at a position

corresponding to the floating region **150** to have a depth for allowing the etching of at least a portion of the floating region **150** and a width  $W'$  narrower than the width  $W$  of the floating region **150**.

(110) In **S60**, a first surface insulating layer **123** may be formed by isotropic-depositing an insulating material (e.g., a silicon oxide) with respect to the entire portion of the surface of the trench **TH**, the well region **118**, and the emitter region **130**.

(111) In **S70**, after an insulating material (e.g., a silicon nitride) is isotropically deposited inside the trench **TH**, an anisotropic etching may be performed to form a protective layer **125**. The protective film **125** may serve as a mask to protect the first surface insulating layer **123** inside the trench **TH** and guide an etching position in a subsequent etching process.

(112) In **S80**, the first surface insulating layer **123** may be etched to a specific depth by employing the protective film **125** as a mask, thereby forming an etched first surface insulating layer **123'**. The depth, in which the first surface insulating layer **123'** is etched, may be equal to the depth from the lower end of the gate electrode **120** to the lower end of the trench emitter region **135**, as illustrated in FIG. 5.

(113) In **S90**, after removing the protective layer **125**, a conductive material (e.g., polysilicon) may be isotropically deposited on the entire portion of the resultant structure to form an electrode layer **120'** to be spaced apart from the surface of the trench **TH**.

(114) In **S100**, the gate electrode **120** may be formed by anisotropic etching the electrode layer **120'**, and the second surface insulating layer **126** may be formed by depositing an insulating material (e.g., a silicon oxide) on the entire portion of the resultant structure.

(115) In **S110**, the conductive material (e.g., polysilicon) is gap-filled inside the trench to form the trench emitter region **135**.

(116) In **S120**, components positioned higher than the upper portion of the emitter region **130** may be removed through a polishing process to planarize the upper portion of the semiconductor substrate **110**. Accordingly, the upper portion of each of the emitter region **130**, the gate electrode **120**, and the trench emitter region **135** may be exposed. According to various embodiments, the emitter region **130** may be formed after planarizing the upper portion of the semiconductor substrate **110** in **S120**, instead of **S40**. The first surface insulating layer **123'** and the second surface insulating layer **126**, in which the polishing process has been finished, may be expressed in the form of one gate insulating film **122'**.

(117) In **S130**, an upper insulating layer **128** may be applied to the upper portion of the semiconductor substrate **110**. The upper insulating layer **128** may have the same material as that of the gate insulating film **122** described above.

(118) In **S140**, an etching mask is disposed on the upper insulating layer **128**, and a vacancy **VC**, which is an empty space, may be formed through the etching process. The position of the etching mask and the etching depth may be determined in advance such that the emitter region **130** may be exposed through the etching process. The emitter region **130** and at least a portion of the well region **118** may be exposed through the etching process. In addition, the gate insulating film **122'** and the upper insulating layer **128** which is etched may have the integral form of the gate insulating film **122** illustrated in FIG. 5A.

(119) In **S150**, as a conductive material is gap-filled in and applied to the upper portion of the gate insulating film **122** and the vacancy **VC** to form the emitter electrode **132**, thereby forming the structure of the power semiconductor device **10** illustrated in FIG. 5A.

(120) According to the power semiconductor device, the power semiconductor chip including the same, and the method for manufacturing the same of an embodiment of the present disclosure, the withstand voltage characteristic may be maintained while the operation stability is improved.

(121) Such an effect is provided only for the illustrative purpose, and embodiments of the present disclosure are not limited thereto.

(122) Various embodiments of the present disclosure do not list all available combinations but are



for describing a representative aspect of the present disclosure, and descriptions of various embodiments may be applied independently or may be applied through a combination of two or more.

(123) A number of embodiments have been described above. Nevertheless, it will be understood that various modifications may be made. For example, suitable results may be achieved if the described techniques are performed in a different order and/or if components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented by other components or their equivalents. Accordingly, other implementations are within the scope of the following claims.

(124) While this disclosure includes specific examples, it will be apparent after an understanding of the disclosure of this application that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

## Claims

1. A power semiconductor device, comprising: a plurality of gate electrodes configured to be recessed from a first surface of a semiconductor substrate to a second surface of the semiconductor substrate, the second surface being opposite to the first surface; an emitter region configured to make contact with a trench and the first surface, wherein the emitter region is provided between respective ones of the plurality of gate electrodes, wherein the emitter region comprises first impurities of a first conductive type; a collector region configured to make contact with the second surface, and comprising second impurities of a second conductive type opposite to the first conductive type; a floating region extending toward the second surface in an extension direction of the trench while surrounding a bottom surface of the trench, and comprising the second impurities; and a trench emitter region interposed between the plurality of gate electrodes in the trench.
2. The power semiconductor device of claim 1, further comprising: a gate insulating film interposed between each of the plurality of gate electrodes and the trench emitter region, and between at least one of each of the plurality of gate electrodes and the floating region, and a surface of the trench.
3. The power semiconductor device of claim 2, further comprising a gate insulating film, the gate insulating film being interposed between a side surface of the trench and the plurality of gate electrodes, the gate insulating film having a first thickness that is less than a second thickness of a gate insulating film interposed between the bottom surface of the trench and the plurality of gate electrodes.
4. The power semiconductor device of claim 1, wherein a first length of the trench emitter region extending from the first surface of the semiconductor substrate is longer than a second length of each of the plurality of gate electrodes extending from the first surface.
5. The power semiconductor device of claim 1, wherein the trench emitter region is electrically connected to the emitter region through an emitter electrode.
6. The power semiconductor device of claim 1, wherein the trench emitter region comprises polysilicon.

7. The power semiconductor device of claim 1, wherein a third length of the floating region in the extension direction is greater than a first width of the floating region.
  8. The power semiconductor device of claim 1, wherein a first width of the floating region is greater than a second width of the trench.
  9. The power semiconductor device of claim 1, wherein a first depth from the first surface to an upper end of the floating region is shallower than a second depth from the first surface to a lower end of the trench.
  10. The power semiconductor device of claim 1, wherein a lower end of the floating region is spaced apart from an upper end of the collector region by a specific distance.
  11. The power semiconductor device of claim 1, wherein the emitter region comprises a plurality of regions disposed at opposite sides of the trench, respectively.
  12. The power semiconductor device of claim 1, further comprising: a drift region disposed on the collector region, the drift region extending in the extension direction while making contact with a side surface of each of the floating region and the trench, wherein the drift region comprises the first impurities.
  13. The power semiconductor device of claim 12, wherein a first doping concentration of the first impurities in the emitter region is heavier than a second doping concentration of the first impurities in the drift region.
  14. The power semiconductor device of claim 1, wherein the plurality of gate electrodes are disposed in a stripe type or a ring type, when viewed from a plan view.
  15. A power semiconductor chip, comprising: a cell region including the power semiconductor device of claim 1; and a peripheral region including a gate terminal, an emitter terminal, and a collector terminal electrically connected to the plurality of gate electrodes, the emitter region, and the collector region, respectively.
  16. A method for manufacturing a power semiconductor device, the method comprising: forming a first drift region comprising first impurities of a first conductive type, through epitaxial growth; forming a floating region by implanting second impurities comprising a second conductive type opposite to the first conductive type, into an upper portion of the first drift region; forming a second drift region comprising the first impurities on the first drift region through the epitaxial growth; forming an emitter region by implanting the first impurities into an upper portion of the second drift region; forming a trench through the emitter region by performing an etching process to etch at least a portion of the floating region; forming a plurality of gate electrodes by depositing a conductive material to be spaced apart from a surface of the trench; and forming a trench emitter region by depositing a conductive material between the plurality of gate electrodes such that the trench emitter region is spaced apart from each of the plurality of gate electrodes.
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